AMERICAN INTERNATIONAL UNIVERSITY BANGLADESH (AIUB)

FACULTY OF SCIENCE & TECHNOLOGY

Course Title

INTRODUCTION TO DATABASE

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Section: I

TITLE

SuperTechIT: Super Shop Database Management System

Supervised By

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CO2: Understand the fundamental concepts underlying database systems and gain hands-on experience with ER diagram Case study		
PO-c2: Develop process for complex computer science and engineering problems considering cultural and societal factors.		Marks

1. Introduction

Technology is now an integral part of our everyday lives, and we depend on various tech products for numerous aspects of our routines. In Bangladesh, it's uncommon to find a household without some technology. In response to this growing reliance, we aim to develop a tech product database system called "**SuperTechIT**" to facilitate seamless customer service. This system is designed to enhance operational efficiency across various sectors, specifically targeting tech product management. By streamlining processes and improving decision-making, SuperTechIT will benefit all stakeholders involved, from management to employees, making their tasks more efficient and effective.

2. Case Study

The **SuperTechIT** management system is designed to streamline operations within a retail environment. It addresses the need to effectively manage customers, products, employees, and companies. In the rapidly growing retail sector, super-shops often face challenges in managing a diverse range of products and customers efficiently. The **SuperTechIT** has been developed to tackle these challenges by providing a structured platform for tracking customer purchases, managing product inventory, and overseeing employee roles. In a **SuperTechIT** Management system, customers can purchase multiple products. Each customer is uniquely identified by their ID, and the system also records their names, phone numbers, and membership status. Products are identified by their ID, name, price, and quantity, and each product is categorized accordingly. Categories are identified by their category ID and name. The system includes employees, with each employee managing multiple customers. Employees are identified by their ID, name, salary, and phone number. Most employees are responsible for restoring various products. Additionally, there are multiple companies, each supplying numerous products, and companies are identified by their ID, name, and phone number.

3. ER Diagram

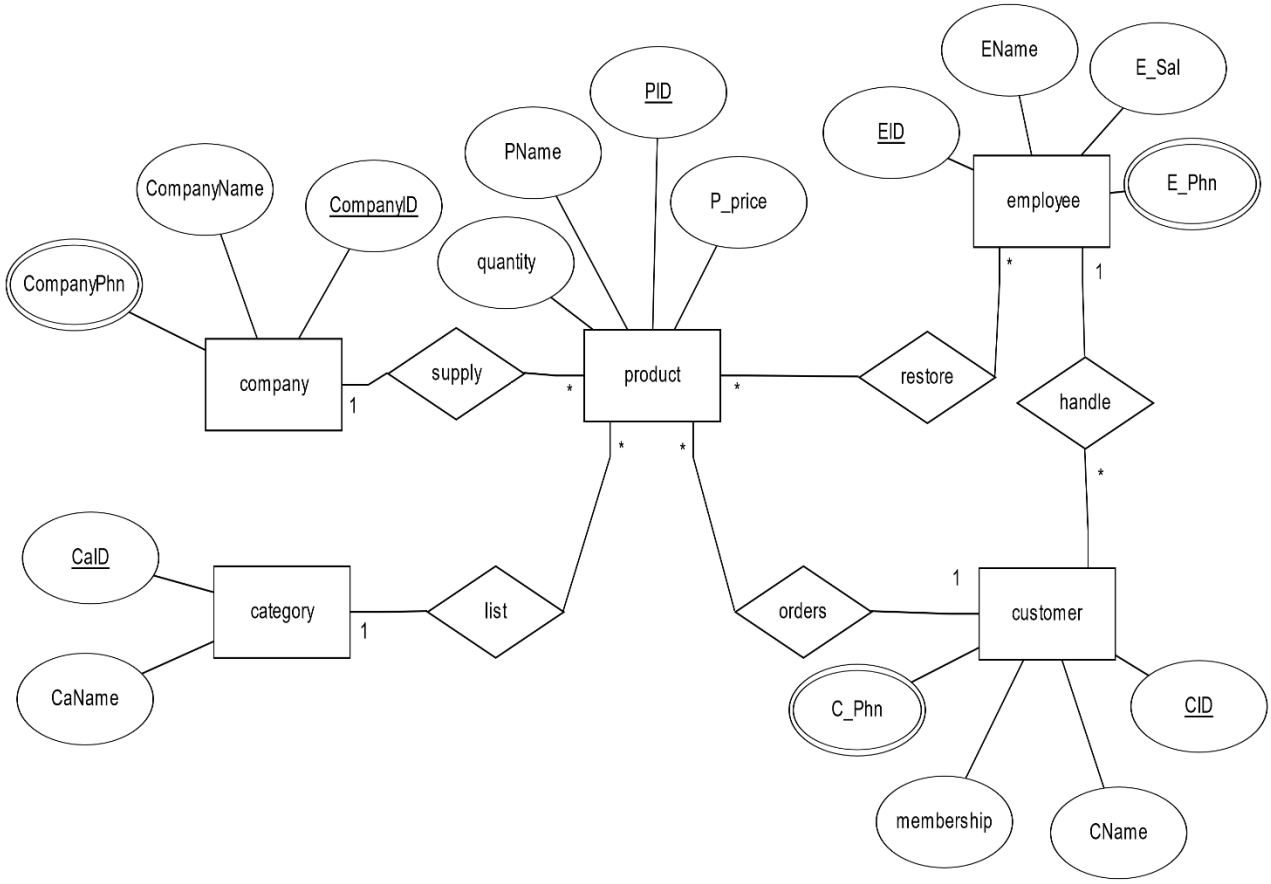


Fig-1: Overall ER Diagram of Our Project

4. Normalization

Relation: One to many

1. supply:

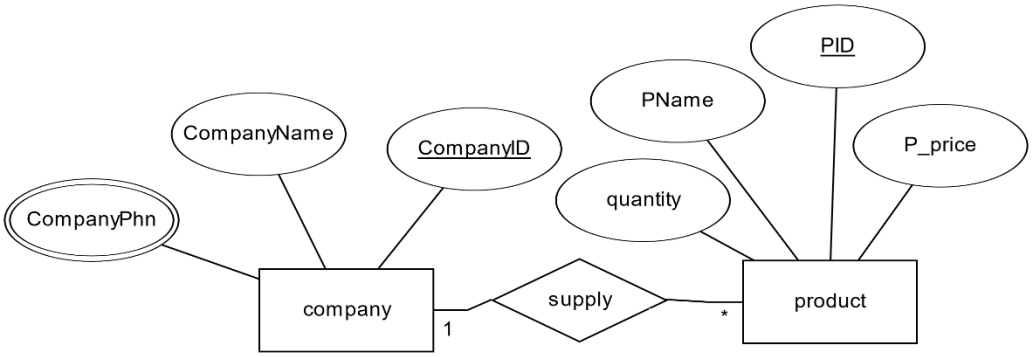


Fig-2: A Company supplies many products

UNF: CompanyID, CompanyName, CompanyPhn, PID, PName, P_price, quantity.

1NF: i) CompanyID, CompanyName, CompanyPhn; ii) PID, PName, P_price, quantity.

2NF:

i) CompanyID (PK), CompanyName, CompanyPhn

ii) PID (PK), PName, P_price, quantity

iii) PID (PK), CompanyID (FK)

iv) CompanyID(FK), CompanyPhn

3NF: There is no transition dependency. 3NF will be same as 2NF.

2. Orders

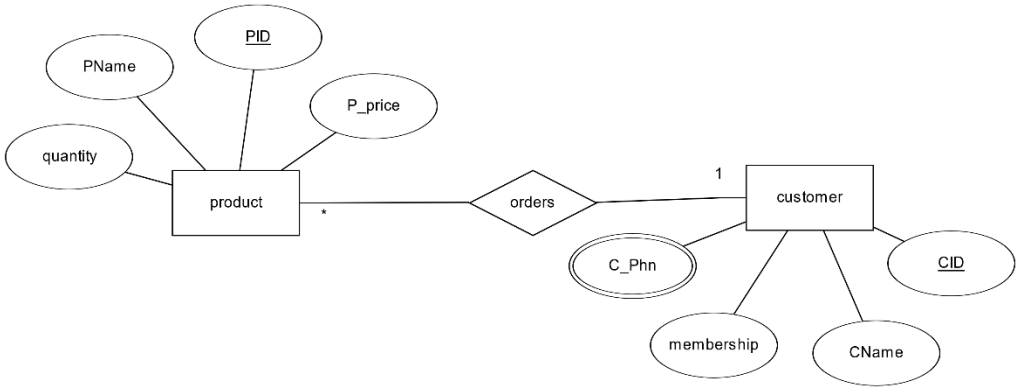


Fig-3: A customer orders many products

UNF: PID, PName, P_price, quantity, CID, CName, C_Phn, membership

1NF: i) PID, PName, P_price, quantity; ii) CID, CName, C_Phn, membership

2NF:

(i) PID (PK), PName, P_price, quantity

(ii) CID (PK), CName, C_Phn, membership

(iii) CID (FK), C_Phn

(iii) PID (PK), CID (FK)

3NF: There is no transition dependency. 3NF will be same as 2NF.

3. handle

UNF: EID, EName, E_Sal, E_Phn, CID, CName, C_Phn, membership

1NF: i) EID, EName, E_Sal, E_Phn;
ii) CID, CName, C_Phn, membership
2NF: i) EID (PK), EName, E_Sal, E_Phn;
ii) EID(FK), E_Phn
iii) CID (PK), CName, C_Phn, membership, EID (FK)
iv) CID (FK), C_Phn
3NF: There is no transition dependency.
3NF will be same as 2NF.

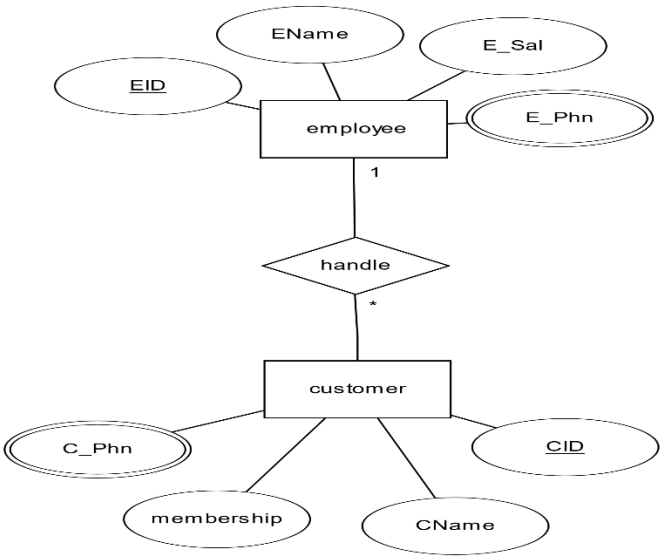


Fig-4: An employee handles many customers

Relation: Many to many

4. restore

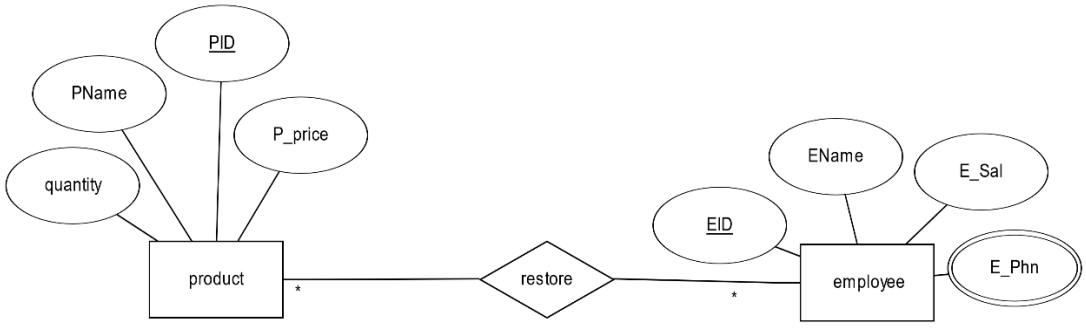


Fig-5: employees restore products

UNF: PID, PName, P_price, quantity, EID, EName, E_Sal, E_Phn

1NF: i) PID, PName, P_price, quantity; ii) EID, EName, E_Sal, E_Phn

2NF: i) PID (PK), PName, P_price, quantity;
ii) EID (PK), EName, E_Sal, E_Phn;
iii) PID (PK), EID (FK)
iv) EID(FK), E_Phn

3NF: Same as 2NF.

Relation: Many to one

5. list

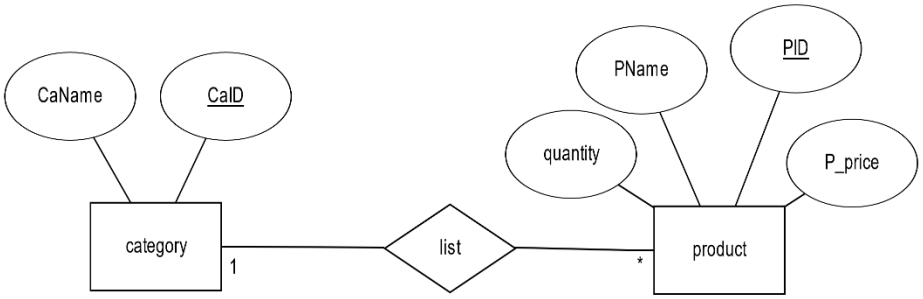


Fig-6: Products list in a category

4. UNF: CaID, CaName, PID, PName, P_price, quantity.

1NF: i) CaID, CaName; ii) PID, PName, P_price, quantity.

2NF: i) CaID (PK), CaName;
ii) PID (PK), PName, P_price, quantity.
iii) PID (PK), CaID (FK)
3NF: Same as 2NF.

5. Finalization

1. EID (PK), EName, E_Phn_Sal, E_Phn **(Employee Table)**
2. PID (PK), PName, P_price, quantity **(Product Table)**
3. CID (PK), CName, C_Phn, membership **(Customer Table)**
4. CID (FK), C-Phn **(Customer_contact)**
5. PID (PK), CID (FK) **(Orders Table)**
6. EID(FK), E_Phn (Employee_Contact
7. CID (PK), CName, membership, EID (FK) **(Handle Table)**
8. PID (PK), EID (FK) **(Restore Table)**
9. CaID (PK), CaName **(Category Table)**
10. PID (PK), CaID (FK) **(List Table)**
11. CompanyID (PK), CompanyName, CompanyPhn **(Company Table)**
12. CompanyID(FK), CompanyPhn **(Company_Info)**
13. PID (PK), CompanyID (FK) **(Supply Table)**